Md Ahsanul Haque, Ph.D.
MAT-120

Homework-1

Date: July 9, 2026

Name	Raihan Ahammed
Section	11
ID	2532600043

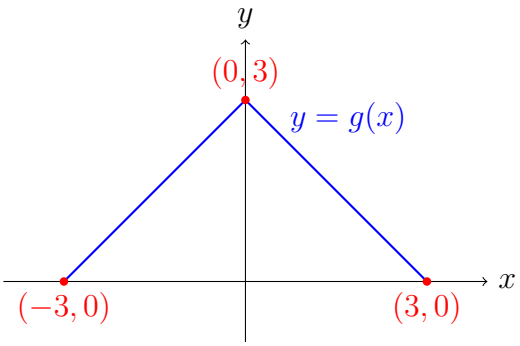
Instructions:

- Write your full name and section number inside the provided box.
- This assignment contains exactly 5 problems.
- Use correct mathematical notation and clearly show all logical steps.

---

1. (10 points)
   (a) Given the function $f(x) = -\sqrt{9 - x^2}$, determine:
      i. The domain and range of $f(x)$.
      ii. Whether $f(x)$ is one-to-one (justify your answer algebraically or graphically).
      iii. The inverse function $f^{-1}(x)$ if it exists, including its restricted domain and range.

   (b) The graph of a piecewise linear function $y = g(x)$ is shown below. Sketch the graphs of the following transformations, clearly labeling the new coordinates of the three key points:



      i. $y = g(x + 1) - 2$
      ii. $y = -\frac{1}{2}g(x)$
      iii. $y = g(-2x)$

2. (10 points)
   (a) Solve for x exactly. Be sure to verify your solutions against the domain of the original equation:

$$\log_3(x+4) - \log_3(x-2) = 1 + \log_3(2)$$

   (b) Given the functions $f(x) = 5\ln(2x-1)$ and $g(x) = e^{x/5}$:
      i. Find the composite function $(f \circ g)(x)$ and state its domain.
      ii. Find the composite function $(g \circ f)(x)$ and state its domain.

3. (10 points) A radioactive isotope decays exponentially according to the model

$$M(t) = M_0 e^{-rt}$$

   where t is measured in days. The isotope has a half-life of 15 days, and the initial sample mass is 800 grams.
   (a) Determine the exact value of the decay constant r.
   (b) How many days will it take for the sample to decay to a mass of exactly 50 grams?
   (c) Find the average rate of decay of the substance between $t = 15$ days and $t = 30$ days. Include units in your answer.

4. (10 points)
   (a) Consider the rational function $f(x) = \dfrac{5x-2}{3x+4}$. Find its inverse function $f^{-1}(x)$ and algebraically verify your result by showing that $f(f^{-1}(x)) = x$.
   (b) Analyze and formally state the symmetry (even, odd, or neither) of the following functions. Show all algebraic testing steps.
      i. $h(x) = x^2 \sin(x)$
      ii. $k(x) = \ln(x^2+1)$

5. (10 points)
   (a) A manufacturing plant models its total weekly cost (in dollars) using the function $C(x) = 500 + 10x + 0.1x^2$, where x represents the number of units produced. The average cost function is defined as $\bar{C}(x) = \frac{C(x)}{x}$.
      i. Find the practical mathematical domain of $\bar{C}(x)$ in the context of manufacturing.
      ii. Write the explicit formula for $\bar{C}(x)$ and evaluate $\bar{C}(10)$. Briefly interpret what this numerical value means for the plant.

   (b) Let $f(x)$ and $g(x)$ both be odd functions that are defined for all real numbers. Let $H(x) = f(x)g(x)$ be their product. Prove algebraically that $H(x)$ must be an even function.

Solution to question no - 1(a)

(i) The given function, $f(x) = -\sqrt{9-x^2}$

Domain,

$9 - x^2 \geq 0$

$\Rightarrow x^2 \leq 9$

$\Rightarrow |x| \leq 3$

$\Rightarrow -3 \leq x \leq 3$

Range,

As, $\sqrt{9-x^2} \geq 0$

$f(x) \leq 0$

Maximum value, when $\sqrt{9-x^2}$ is minimum,

$\therefore 9 - x^2 = 0 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$

$\therefore f(\pm 3) = 0$

Minimum value, when $\sqrt{9-x^2}$ is maximum

If, $x = 0$ then,

$\sqrt{9-0} = 3$

$\therefore f(0) = -3$

$\therefore$ Domain : $-3 \leq x \leq 3$ or, $[-3, 3]$

Range : $-3 \leq y \leq 0$ or, $[-3, 0]$

(ii) The given function, $f(x) = -\sqrt{9-x^2}$

For, $\pm x$; $x^2 \geq 0$

$\therefore$ Under the domain $[-3, 3]$,

$f(x) = -\sqrt{9-x^2}$

and,

$f(-x) = -\sqrt{9-(-x)^2} = f(x) - \sqrt{9-x^2} = f(x)$

$\therefore$ Different input gives same output.

$\therefore f(x) = -\sqrt{9-x^2}$ is not one-to-one

(iii) As $f(x) = -\sqrt{9-x^2}$ is not one-to-one,

$\therefore f^{-1}(x)$ does not exist.

~~If we~~
So, we restrict the domain: $-3 \leq x \leq 3$

let, $0 \leq x \leq 3$

$\therefore f(x) = y = -\sqrt{9-x^2}$

$\Rightarrow y^2 = 9 - x^2$

$\Rightarrow x^2 = 9 - y^2$

$\Rightarrow x = \pm\sqrt{9-y^2}$

$\therefore x = \sqrt{9-y^2}$ $[\because 0 \leq x \leq 3]$

$\therefore f^{-1}(x) = \sqrt{9-x^2}$

Solution to question no. – 1 (b)

(i)

For, $y = g(x)$, three coordinates are, $(-3,0), (0,3), (3,0)$

if $y = g(x+1) - 2$

$(x+1)$ → horizontal shift : 1 unit left

-2 → vertical shift : 2 unit down

∴ New coordinates,

$(-3,0) \rightarrow x+1 = -3 \quad y = 0-2$

$\therefore x = -4 \quad \therefore y = -2$

$\therefore (-4, -2)$

$(0,3) \rightarrow x+1 = 0 \quad y = 3-2$

$\Rightarrow x = -1 \quad \Rightarrow y = 1$

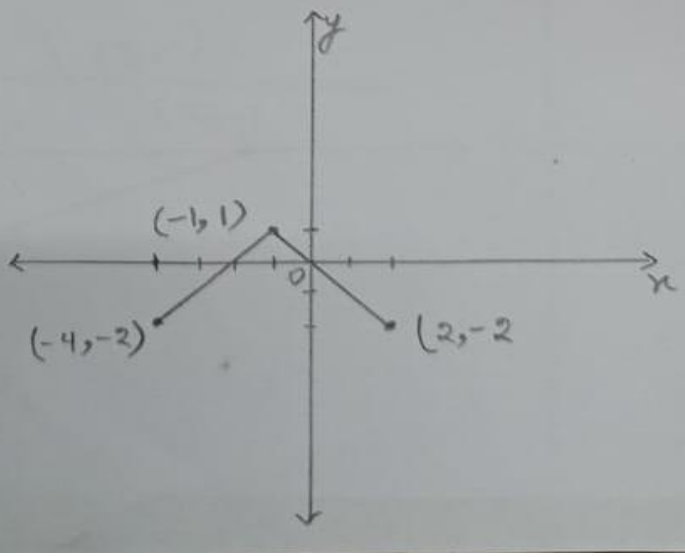
$\therefore (-1, 1)$

$(3,0) \rightarrow x+1 = 3 \quad y = 0-2$

$\Rightarrow x = 2 \quad \Rightarrow y = -2$

$\therefore (2, -2)$

∴ New coordinates :
$(-4, -2), (-1, 1), (2, -2)$



(ii)

~~g(x)~~ For, $y = 2g(x)$; three coordinates : $(0,3), (-3,0), (3,0)$

if $y = \frac{-1}{2} g(x)$

$-g(x) \rightarrow$ reflect about x-axis

$\frac{1}{2} g(x) \rightarrow$ compress vertically

∴ New points,

$(-3,0) \rightarrow x : -3 \quad y : \frac{-1}{2} \times 0 = 0$

$\therefore (x,y) = (-3,0)$

$(0,3) \rightarrow x : 0 \quad y : -\frac{1}{2} \times 3 = -\frac{3}{2}$

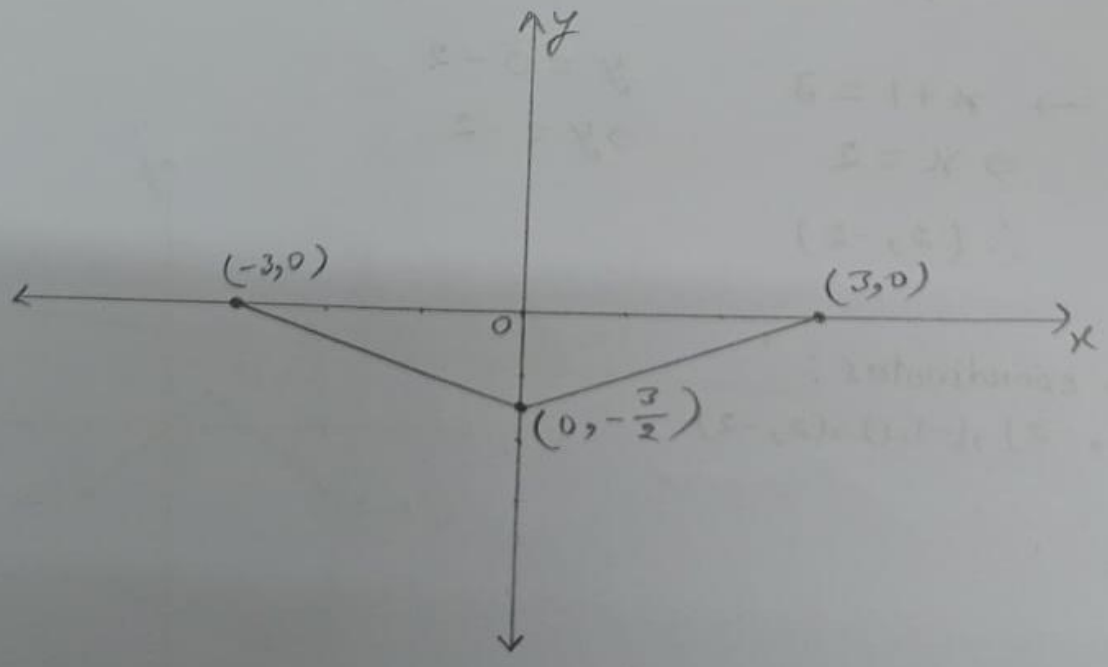
$\therefore (x,y) = (0, -\frac{3}{2})$

$(3,0) \rightarrow x : 3 \quad y : \frac{-1}{2} \times 0 = 0$

$\therefore (x,y) = (3,0)$

∴ New points,

$(-3,0), (0, -\frac{3}{2}), (3,0)$



(iii)

The three points : $(-3,0),(0,3),(3,0)$

For, $y = g(-2x)$

$g(-x) \rightarrow$ reflection about y-axis

$g(2x) \rightarrow$ horizontal compression

$\therefore$ New points,

$(-3,0) \rightarrow x : -2x = -3 \qquad y : 0$

$\therefore x = \frac{3}{2}$

$(0,3) \rightarrow x : -2x = 0 \qquad y : 3$

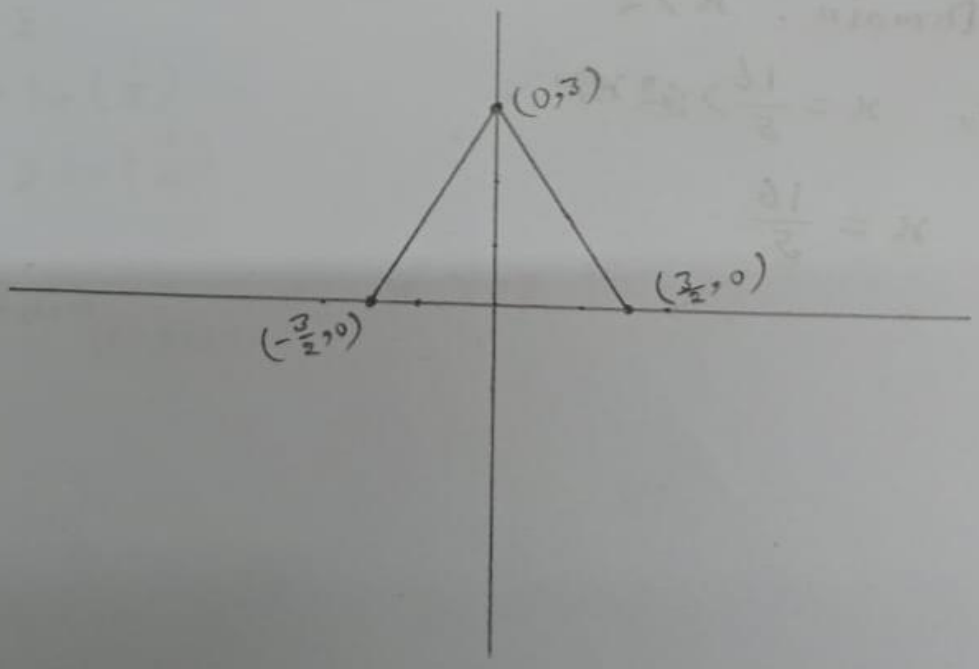
$\therefore x = 0$

$(3,0) \rightarrow x : -2x = 3 \qquad y : 0$

$\therefore x = \frac{-3}{2}$

$\therefore$ New points are,

$(\frac{3}{2},0),(0,3),(\frac{-3}{2},0)$



Solution to question - 2(a)

given equation: $\log_3(x+4) - \log_3(x-2) = 1 + \log_3 2$

$$\Rightarrow \log_3 \frac{x+4}{x-2} = \log_3 3 + \log_3 2$$

$$\Rightarrow \log_3 \frac{x+4}{x-2} = \log_3 6$$

$$\Rightarrow \frac{x+4}{x-2} = 6$$

$$\Rightarrow 6x - 12 = x + 4$$

$$\Rightarrow x = \frac{16}{5}$$

To verify the solution,

domain of log, $x+4>0$ and $x-2>0$

$\therefore x > -4$ and $x > 2$

$x > -4 \cap x > 2$

$\therefore x > 2$

$\therefore$ Domain : $x > 2$

As, $x = \frac{16}{5} > 2$

$\therefore x = \frac{16}{5}$

Solution to question no.-2 (b)

Given function, $f(x) = 5\ln(2x-1)$ and $g(x) = e^{x/5}$

(i)

$$(f \circ g)(x) = f(g(x))$$
$$= f(e^{x/5})$$
$$= 5\ln(2e^{x/5} - 1)$$

Domain of $g(x)$: $x \in \mathbb{R}$

Domain of $f(x)$,

$2x - 1 > 0$

$\Rightarrow x > \frac{1}{2}$

$\therefore \text{In } (f \circ g)(x), \quad g(x) > \frac{1}{2}$

$\therefore$ Domain of $(f \circ g)(x)$,

$e^{x/5} > \frac{1}{2}$

$\Rightarrow \frac{x}{5} > \ln\left(\frac{1}{2}\right)$

$\therefore x > 5\ln\left(\frac{1}{2}\right)$

$\therefore \text{Domain}_{(f \circ g)(x)} : \quad x > 5\ln\left(\frac{1}{2}\right)$

(ii)

$$(g \circ f)(x) = g(f(x))$$
$$= g(5\ln(2x-1))$$
$$= e^{\frac{5\ln(2x-1)}{5}}$$
$$= e^{\ln(2x-1)}$$
$$= 2x - 1$$

Domain of $f(x)$,

$2x - 1 > 0$

$\Rightarrow 2x > 1$

$\therefore x > \frac{1}{2}$

Domain of $g(x)$: $x \in \mathbb{R}$

$\therefore$ Domain $_{(g \circ f)(x)}$: $x > \frac{1}{2}$

Solution to question no. – 3

a) Decay model : $M(t) = M_0 e^{-rt}$

Initial mass $M_0(t) = 800$

After 15 days, $M = \frac{M_0}{2}$

$\therefore M(15) = M_0 e^{-r \cdot 15}$

$\Rightarrow \frac{M_0}{2} = M_0 e^{-r \cdot 15}$

$\Rightarrow e^{-15r} = \frac{1}{2}$

$\Rightarrow -15r = \ln\left(\frac{1}{2}\right)$

$\Rightarrow 15r = \ln 2$

$\therefore r = \frac{\ln 2}{15}$

$\therefore$ Exact value of decay constant, $r = \frac{\ln 2}{15}$

b) $M(t) = M_0 e^{-rt}$; $M_0 = 800\ g$, $r = \frac{\ln 2}{15}$

To decay to 50 g.

$M(t) = 50 = M_0 e^{-rt}$

$\Rightarrow 50 = 800\, e^{-rt}$

$\Rightarrow \frac{1}{16} = e^{-rt}$

$\Rightarrow \ln\left(\frac{1}{16}\right) = \ln e^{-rt}$

$\Rightarrow -\ln 16 = -rt$

$\therefore t = \frac{\ln 16}{r} = \frac{\ln 16}{\frac{\ln 2}{15}} = \frac{15 \ln 16}{\ln 2} = 60$ days

$\therefore$ 60 days need to decay to 50 g.

c)

$M(t) = M_0 e^{-rt}$; $M_0 = 800\ g$, $r = \frac{\ln 2}{15}$

For, $t = 15$ day

Mass $M(15) = M_0 e^{-r15}$

$= 800\, e^{-\frac{(\ln 2)\times 15}{15}}$

$= \frac{800}{2}$

$= 400\ g$

For, $t = 30$ days,

Mass $M(30) = M_0 e^{-30r}$

$= 800\, e^{-\frac{30\times \ln 2}{15}}$

$= 800\, e^{-2\ln 2}$

$= 800 \times \frac{1}{4}$

$= 200\ g$

$\therefore$ Average rate of decay, $\frac{M(30) - M(15)\ g}{(30-15)\ day}$

$= \frac{200 - 400}{15}\ g/day$

$= -\frac{40}{3}\ g/day$

$\therefore$ Average rate of decay is $-\frac{40}{3}$ or, $-13.33\ g/day$

Solution to question no. - 4

a) given function, $f(x) = \frac{5x-2}{3x+4}$

$$y = \frac{5x-2}{3x+4}$$

$$\Rightarrow y(3x+4) = 5x-2$$

$$\Rightarrow 3xy + 4y = 5x - 2$$

$$\Rightarrow 3xy - 5x = -2 - 4y$$

$$\Rightarrow x(3y-5) = -(4y+2)$$

$$\therefore x = \frac{4y+2}{5-3y}$$

$$\therefore f^{-1}(x) = \frac{4x+2}{5-3x}$$

Now, $f(f^{-1}(x)) = f\left(\frac{4x+2}{5-3x}\right)$

$$= \frac{5\left(\frac{4x+2}{5-3x}\right) - 2}{3\left(\frac{4x+2}{5-3x}\right) + 4}$$

$$= \frac{\frac{20x+10-10+6x}{5-3x}}{\frac{12x+6+20-12x}{5-3x}}$$

$$= \frac{26x}{26}$$

$$= x$$

$$\therefore f(f^{-1}(x)) = x$$

$$\therefore f^{-1}(x) = \frac{4x+2}{5-3x}$$

(4. b)

i) Given function, $h(x) = x^2 \sin(x)$

To find if its even or odd or neither,

$$h(-x) = (-x)^2 \sin(-x)$$
$$= x^2 \sin(-x)$$
$$= -x^2 \sin(x) \quad [\because \sin(-x) = -\sin x]$$
$$= -h(x)$$

As, $h(-x) = -h(x)$

$\therefore h(x) = x^2 \sin(x)$ is odd function.

ii) $k(x) = \ln(x^2+1)$

Now,

$$k(-x) = \ln((-x)^2 + 1)$$
$$= \ln(x^2+1)$$
$$= k(x)$$

As, $k(-x) = k(x)$

$\therefore k(x) = \ln(x^2+1)$ is a even function.

Solution to question – 5 (a)

(i)

Weekly cost function : $C(x) = 500 + 10x + 0.1x^2$

Average cost $\bar{C}(x) = \frac{C(x)}{x}$

$= \frac{500 + 10x + 0.1x^2}{x}$

domain of $\bar{C}(x)$,

$x \neq 0$

$x > 0$ [∵ no negative no. of units can be produced]

$x \neq 0 \cap x > 0$

$\therefore x > 0$

∴ Practical domain is $x > 0$.

(ii)

$\bar{C}(x) = \frac{C(x)}{x}$; x = number of unit produced

$= \frac{500 + 10x + 0.1x^2}{x}$

$= \frac{500}{x} + 10 + 0.1x$

$\therefore \bar{C}(10) = \frac{500}{10} + 10 + 0.1 \times 10$

$= 50 + 10 + 1$

$= 61$ dollars

∴ 61 Dollar is the average cost to produce 10 units of product.

If the plant produces 10 unit of product the average cost is 61$.

Solution to question no. 05(b)

$f(x)$ and $g(x)$ are odd function

$\therefore f(-x) = -f(x)$

$g(-x) = -g(x)$

Now,

$H(x) = f(x)g(x)$

and,

$H(-x) = f(-x)\, g(-x)$

$= \{-f(x)\}\{g(-x)\}$

$= f(x)\, g(x)$

$= H(x)$

Since, $H(-x) = H(x)$

$\therefore H(x)$ is an even function.